

# Catch every strip defect where it is born, not where it is shipped

AI surface inspection that finds, grades and routes every stainless strip defect to the process that caused it.

90%

of defects caught on real mill strip it had never seen

4.7%

of clean strip wrongly flagged, fewer than 1 in 20

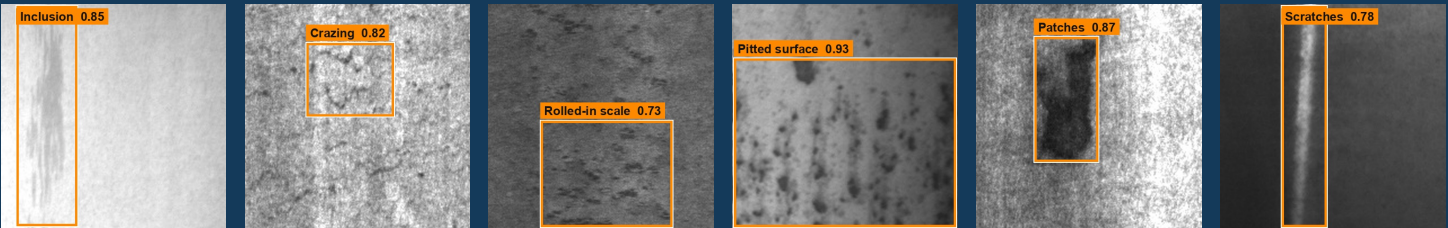
₹2.5-4.2 cr

per line, vs ₹10-29 cr for a commercial system

~2.2 years

payback on one line at a 1% downgrade rate

REAL DETECTIONS BY OUR MODEL, ON STEEL IMAGES IT NEVER SAW IN TRAINING



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TRY THE LIVE DEMO

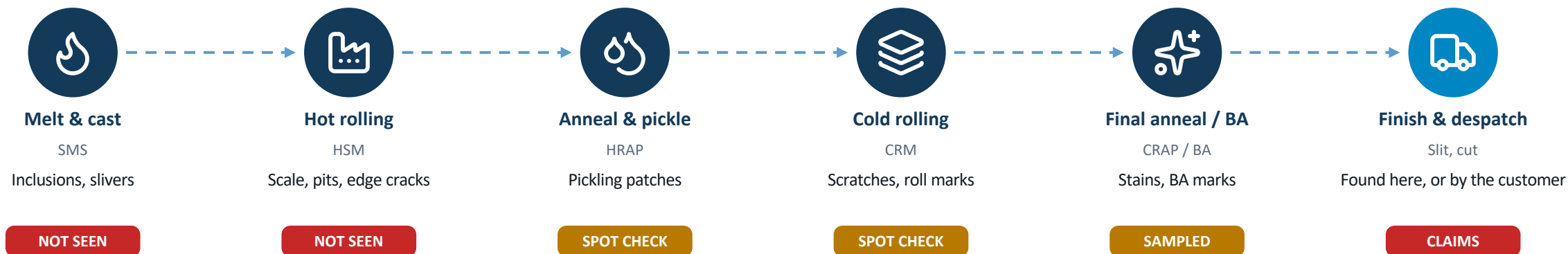
[surface-vision.github.io](https://surface-vision.github.io)

Runs in your browser. Nothing leaves the page.

# A defect born at the caster is paid for at every later stage, but today it is found only at the end

Where stainless surface defects are born, and where they are seen today

VALUE ADDED KEEPS RISING →



## Why it hurts more on stainless

- 300 m/min**  
too fast for the eye to see the whole surface
- ₹126k-165k**  
of nickel in every tonne of 304, given away on a downgrade
- No repair**  
damage on 2B/BA mill finishes cannot be polished out

## Why now

- 4.8 → 6.8 Mt**  
India stainless demand, FY25 to FY30; rail and metro coaches are stainless
- 1.73 Mt**  
imports in FY25; anti-dumping probe on cold-rolled 300/400 series, Sep 2025
- IS 6911**  
BIS certification mandatory for stainless sheet and strip since Aug 2024
- 2.67 MTPA**  
JSL cold rolling by FY28, up from 2.05: more surface-critical tonnes

## Cost of quality: where AI moves the money

← spend moves from failure to prevention

|                                                          |                                                              |
|----------------------------------------------------------|--------------------------------------------------------------|
| <b>Prevention</b><br>defect sent to its owner in minutes | <b>Internal failure</b><br>caught before more value is added |
| <b>Appraisal</b><br>100% of the surface, every coil      | <b>External failure</b><br>fewer escapes to customers        |

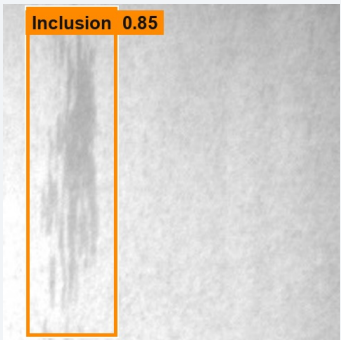
Poor quality costs ~15% of sales in manufacturing (IISE).

Sources: AMETEK/Ternium (2020); ASSDA; IISE; ASTM A240; LME nickel USD 16,402/t at INR 95.8/USD (22 Sep 2026); ISSDA (FY25); DGTR (29 Sep 2025); BIS Steel QCO 2024; JSL.

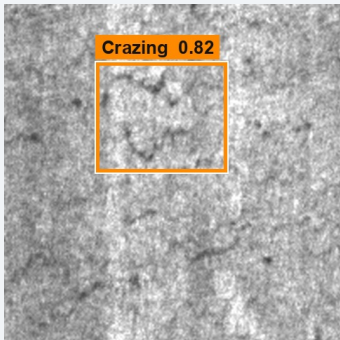
The problem is not that defects exist. They are found late, by sampling, and nobody is told which process caused them.

On stainless the surface is the product, so every defect type must reach the process that caused it

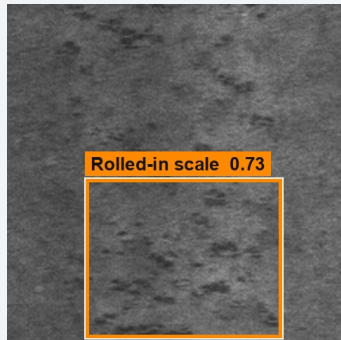
Six defect families, six different owners (real detections by our model)



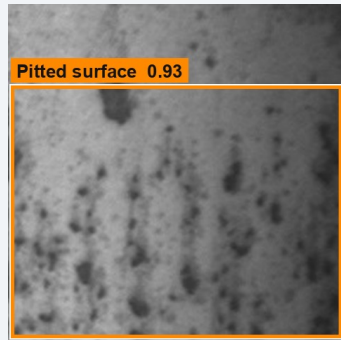
Born: **Caster**  
Owner: **SMS**  
Fix: Tundish level, mould flux



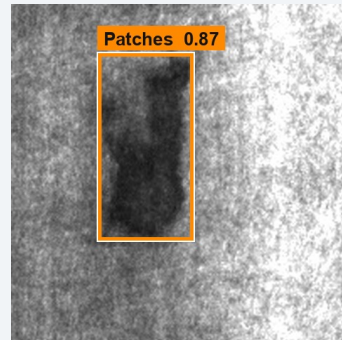
Born: **Reheat, hot rolling**  
Owner: **HSM**  
Fix: Reheat and coiling temperature



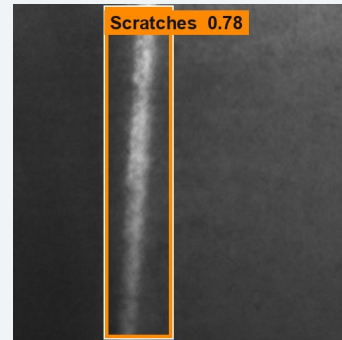
Born: **Hot rolling**  
Owner: **HSM**  
Fix: Descaler pressure, nozzles



Born: **Reheat, pickling**  
Owner: **HSM / pickling**  
Fix: Furnace atmosphere, over-pickling



Born: **Anneal & pickle**  
Owner: **Pickling line**  
Fix: Acid strength, line speed



Born: **Cold rolling, handling**  
Owner: **CRM / logistics**  
Fix: Guides, rolls, coil handling

Precedent: who already does this, and what we copy

|            |                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------|
| Outokumpu  | Europe's #1 stainless maker; AI inspection live at Tornio since 2024, runs offline. <b>Copy:</b> offline edge, multi-site |
| Tata Steel | Kalinganagar (WEF Lighthouse) finds cold-rolled surface defects by video analytics. <b>Copy:</b> in-house model team      |
| Parsytec   | On stainless lines: 170 µm cameras, bright + dark-field light, a decision per coil. <b>Copy:</b> optics, coil decision    |

Grade-by-defect risk: which defects to train for first

| Grade                 | Slivers | Scale | Scratches | Edge cracks | Ridging |
|-----------------------|---------|-------|-----------|-------------|---------|
| 200 series (Cr-Mn)    | M       | M     | H         | M           | L       |
| 300 series (304, 316) | H       | M     | H         | L           | L       |
| 400 series (409, 430) | H       | M     | M         | M           | H       |
| Duplex (2205)         | M       | H     | M         | H           | L       |

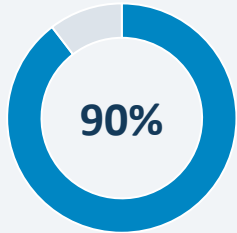
**OUTPUT**  
300 series is the pilot grade: highest exposed-finish risk and the most nickel per tonne.

Images: held-out test frames, boxes by our model. Sources: Nortal/Outokumpu (2024); Tata Steel; Parsytec (Stainless Steel World); Leão 2021; Steel in Translation 2023. Owners, grade risks: team hypotheses.

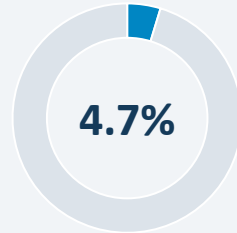
Classify, don't just alarm: a typed defect becomes a work order for a named owner. The pilot belongs on 300-series cold-rolled strip.

## Four design choices behind the system, each proven in our own tests

### 1 Trained on clean steel too, so its alarms mean something



of defects caught  
258 of 288 defective mill  
images



of clean strip flagged  
14 of 300 clean mill  
images

**So we:** train on JSL's own clean and defective strip before any alarm is trusted.

### 2 A compact model is enough

**12 MB**

the whole model; loads in any  
browser

**~55 ms**

per image on an ordinary laptop  
CPU

**0.75**

accuracy, inside the 0.70-0.80  
published range

*Larger models and sharper images added no accuracy in our tests.*

**So we:** keep the model compact and spend the budget on camera coverage.

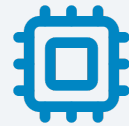
### 3 Line speed, not the model, sets the hardware bill



**~18 processors**

see every 0.2 mm of a 1.28 m strip at 250 m/min

vs



**1 processor**

if images are coarsened to 1.28 mm, but fine  
scratches vanish

**So we:** fix the image resolution with JSL first, then size the hardware.

### 4 Operators act only on alarms they can trust



**±3 pts**

between stated confidence and reality, so a "70%" alarm is right  
about 70% of the time



**0 of 3,200**

real steel images wrongly rejected by our "is this steel?" check,  
which stops the system judging anything else

**So we:** every alarm carries a trustworthy confidence and a steel check.

All numbers measured by us on three public steel-surface datasets, on images the model never saw in training.

**The model is the cheap part. JSL's own strip, the image resolution and trustworthy confidence decide whether the system works.**

# Detect on the strip, decide at the coil, route to the owner, and learn from every correction

## How it works: six stages from camera to corrective action



### Capture

Both faces, bright + dark-field light, 0.2 mm detail

TO BUILD



### Check

Is it usable steel? If not, skip it

BUILT



### Detect

Find and classify every defect

BUILT



### Score

Confidence, severity 0-100, coil position

BUILT



### Decide

Accept, downgrade or hold, per grade

PARTIAL

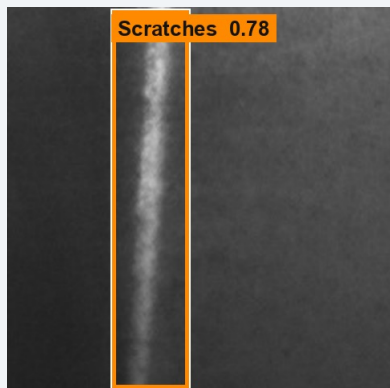


### Act & learn

Alert, hold, work order; corrections retrain it

TO BUILD

## Running today



### Upload a strip image, get:

- box on the defect
- defect type
- confidence
- severity score

[surface-vision.github.io](https://surface-vision.github.io) runs in the browser

## What is new beyond a standard detector



### Trained on clean strip too

only 4.7% of clean strip raises an alarm



### Root-cause routing

every alarm becomes a work order



### Unseen-defect check

flags patterns never labelled (proposed)



### Learns from unlabelled coils

JSL's image archive becomes an asset

## Build vs buy

|                                                   | Cost per line | Proven at speed | Tuned to JSL grades | JSL owns model |
|---------------------------------------------------|---------------|-----------------|---------------------|----------------|
| <b>Commercial</b><br>USD 1-3M (unverified)        | ●             | ●               | ●                   | ●              |
| <b>Fully in-house</b><br>₹2-3 cr (est.)           | ●             | ●               | ●                   | ●              |
| <b>Hybrid (recommended)</b><br>₹2.5-4.2 cr (est.) | ●             | ●               | ●                   | ●              |

● strong ● partial ● weak

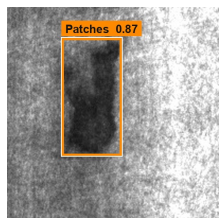
**Output: hybrid is never weak.**

BUILT = working in our prototype today, live at [surface-vision.github.io](https://surface-vision.github.io). Unlabelled-learning result: J. Mater. Inf. 2025. Commercial price range: industry figure, unverified.

**Buy proven cameras, own the model: vendor-grade capture with a detector trained on JSL's grades and wired to JSL's process owners.**

## It already works on public steel data; the pilot's job is to prove it on JSL's own strip

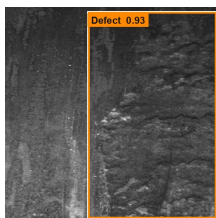
### Tested on three public steel datasets



#### NEU-DET

1,800 images, 6 defect types

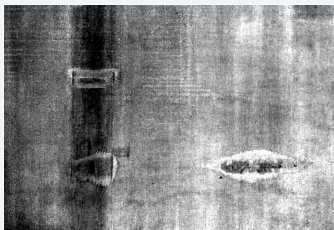
University benchmark, hot-rolled strip



#### Severstal

12,568 real mill frames

Production strip from a steel mill



#### GC10-DET

2,294 frames, 10 types

Full-width line-scan camera frames (roll marks shown)

### How well it finds each defect type (accuracy, held-out images)



published range

### Headline results



178 of 180

test images: the right defect type named



~55 ms

per image, running in an ordinary laptop browser



~10,100

images across three datasets, for training and testing

### Known weak spots, and the fix for each

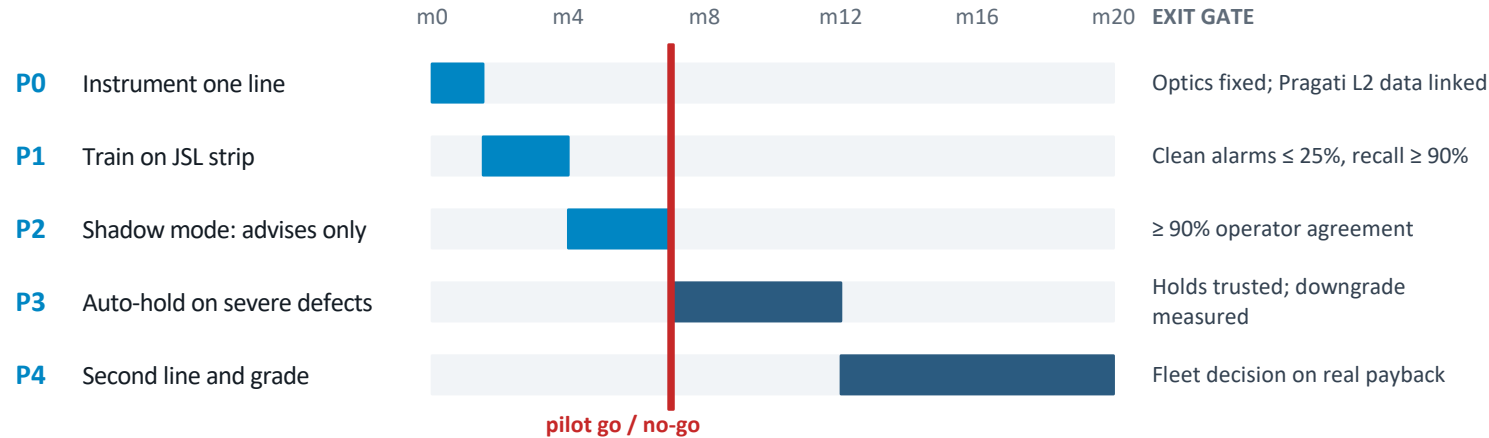
| Weak spot                                          | Today (0-1)  | Fix in the pilot                                 |
|----------------------------------------------------|--------------|--------------------------------------------------|
| Low-contrast texture: crazing, rolled-in scale     | 0.58-0.60    | Angled lighting; JSL labels in P1                |
| Real mill frames are harder than the lab benchmark | 0.57 vs 0.75 | Train on JSL's own strip in P1                   |
| Roll marks and edge cracks: few examples           | 0.20         | Edge cameras; synthetic defects; targeted labels |

Datasets: NEU-DET (Northeastern Univ.); Severstal steel defects (Kaggle); GC10-DET (Lv et al. 2020, CC BY 4.0). All scores on held-out images the model never saw in training; model chosen on validation across four training runs.

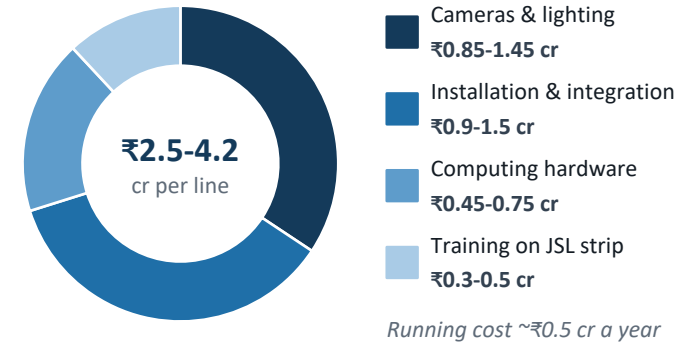
Strong on four of six defect families today. Every weak spot is known, and each has a fix in the pilot plan.

## Seven months to shadow mode on one line, and nothing touches the mill until trust is earned

### Stage-gate roadmap (months)



### Cost per line, estimated



### Top risks, and when each is retired

| Risk                                   | Mitigation                                       | Retired |
|----------------------------------------|--------------------------------------------------|---------|
| Model does not transfer to JSL strip   | Train on JSL clean and defect images             | P1      |
| Mirror-like BA/2B finish hides defects | Bright + dark-field light; deflectometry for BA  | P0-P1   |
| Rare defects under-learnt              | Edge cameras; synthetic defects; targeted labels | P1-P3   |
| Operators ignore alarms                | Trusted scores; shadow mode first                | P2      |
| Plant network, heat and fumes          | IEC 62443 zones, runs offline; sealed enclosures | P0      |
| Quality slips after a model update     | Every retrain must beat the live model first     | P2+     |

### Pilot KPIs: how JSL will know it works

 **$\geq 90\%$** defective strip flagged  
measured 258 of 288 (89.6%) **$\leq 25\%$** clean strip wrongly flagged  
measured 4.7% on mill strip **$\leq 5$  pts**confidence vs reality gap  
measured 3.3 pts **$< 0.2$  s**defect to operator screen  
design target **$\geq 90\%$** operator agreement, shadow  
measured in P2 **$< 8$  h**retrain for a new grade  
measured 0.6 h

Costs: team estimates for a 1.3-1.6 m line, both faces, at INR 95.8/USD; vendor quotes replace them in P0. Pragati: JSL digitalisation, Phase 2 brings Level-2 data at Jajpur.

**Commit ₹2.5-4.2 cr and one line for seven months. The month-7 go/no-go is decided on JSL's own alarm and recall numbers.**

## At a conservative 1% downgrade rate one line pays back in ~2.2 years; the fleet protects ~₹13 cr a year

### Loss pool per line, ₹ cr a year

$0.4 \text{ Mt} \times \text{downgrade rate} \times \text{price discount} \times ₹167,407/\text{t}$

| Downgraded     | 10% off | 15% off | 20% off |
|----------------|---------|---------|---------|
| 0.5% of volume | 3.3     | 5.0     | 6.7     |
| 1.0% of volume | 6.7     | 10.0    | 13.4    |
| 2.0% of volume | 13.4    | 20.1    | 26.8    |

**Base case (dark):** ₹6.7 cr pool, ₹2.0 cr/yr recovered (₹1.5 cr net). Fleet: ₹12.9 cr/yr. Non-prime coil sells 10-30% below prime, so these discounts are conservative.

### Payback per line, base case

#### Our solution

**2.2 yr**

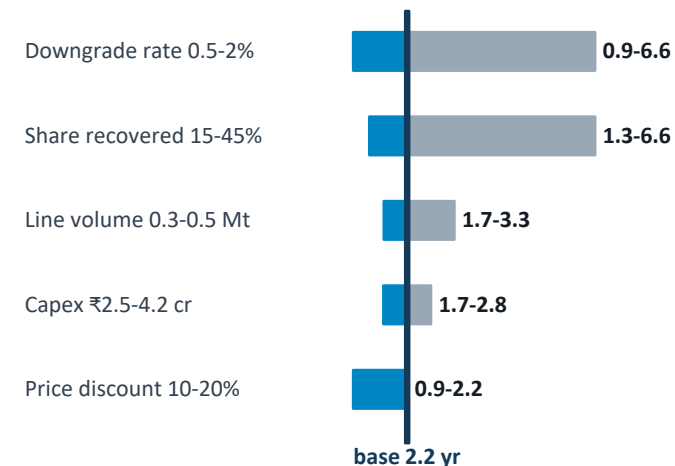
#### Commercial system

**12.7 yr**

**+₹2.1 cr** 5-year NPV per line at 12%

**≥ 0.11%** of line revenue lost to downgrade is enough for a 2-year payback

### What moves payback (years)



### Impact beyond the P&L



#### Coverage

100% of both faces, every coil



#### Customers

Defect map per coil for rail, metro, auto buyers



#### Process

Defects tied to heat and coil IDs



#### Safety

Severe-defect holds prevent strip breaks



#### Sustainability

Less rework and remelt; backs JSL net zero



#### People

Inspectors move to decisions

### WHAT WE ASK OF JSL

- One 300-series cold-rolled line, 7 months
- ₹2.5-4.2 cr pilot budget
- Downgrade rate by line and grade
- A metallurgist at 20% for root causes

**In return: measured downgrade rate, payback and a go / no-go at month 7.**

Inputs: JSL FY26 (INR 167,407/t; 2,565,902 t). Base: 0.4 Mt line, 30% of loss recovered, ₹3.35 cr capex, ₹0.5 cr/yr run cost, 12% rate. Commercial: USD 2M at INR 95.8.

**Recommendation: approve a 7-month pilot on one 300-series cold-rolled line. JSL's own downgrade rate decides fleet rollout.**